

DATA VORTEX

Round 2 — Evaluation Metrics Report

Social Engine NLP: Sentiment Classification and Topic Categorization

Purpose

This report consolidates the baseline evaluation metrics recorded in the submitted project archive and the improved-model results provided by the team in the conversation. The improved results are reported as supplied; the archive's JSON metrics represent the baseline results. These two sources are kept distinct.

Evaluation set

The archived metric files record 1,789 evaluation rows for each task. Metrics are shown as proportions (0–1), with rounded values in tables.

1. Executive Summary

Task	Baseline accuracy	Improved accuracy	Improved macro F1	Key observation
Sentiment	0.592	0.615	0.610	Modest gains across all three sentiment classes.
Topic	0.906	0.949	0.720	Strong aggregate accuracy; rare-class recall remains limited.

*Improved metrics are taken from the team-provided classification reports. Confirm that these correspond to the final, held-out evaluation before submission.

2. Sentiment Classification

Class	Precision	Recall	F1	Support
Negative	0.62	0.65	0.63	597
Neutral	0.51	0.52	0.52	590
Positive	0.65	0.60	0.63	602
macro avg	0.59	0.59	0.59	1789
weighted avg	0.59	0.59	0.59	1789
Macro average	0.59	0.59	0.59	1789
Weighted average	0.59	0.59	0.59	1789

Improved sentiment results supplied by the team: accuracy 0.6149; Negative F1 0.66, Neutral F1 0.53, Positive F1 0.65; macro F1 approximately 0.61. The validation tuning table separately showed C=1.0 with accuracy 0.6757 and macro F1 0.6748; this is a validation result, not interchangeable with the reported test result.

3. Topic Classification

Class	Precision	Recall	F1	Support
Account_Security	0.62	0.31	0.41	26
Community_Discussion	0.91	0.99	0.95	1544
Feature_Feedback	0.44	0.11	0.17	74
Technical_Issues	0.89	0.58	0.70	145
macro avg	0.72	0.50	0.56	1789
weighted avg	0.89	0.91	0.89	1789
Macro average	0.72	0.50	0.56	1789
Weighted average	0.89	0.91	0.89	1789

Improved topic results supplied by the team: accuracy 0.9491, macro F1 0.72, weighted F1 0.94. Per-class F1: Account_Security 0.51, Community_Discussion 0.97, Feature_Feedback 0.48, Technical_Issues 0.92.

4. Confusion Matrices

The following figures are the confusion-matrix image files included in the project archive. Verify that they correspond to the intended final evaluation run; filenames alone do not establish model version.

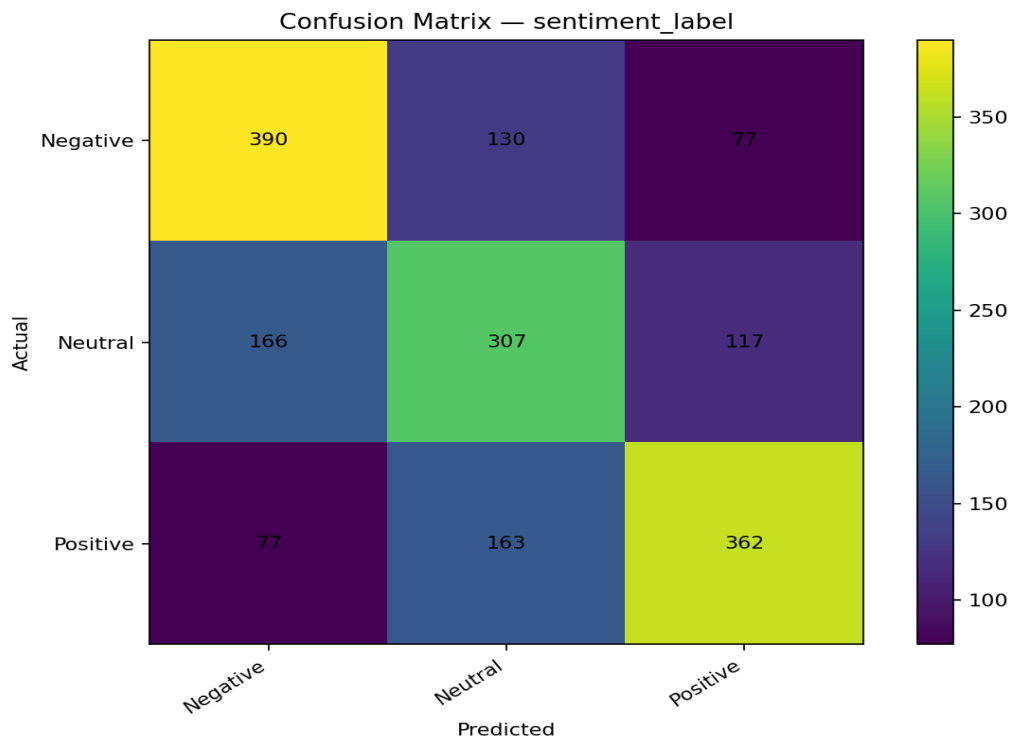


Figure 1. Sentiment confusion matrix (archived output).

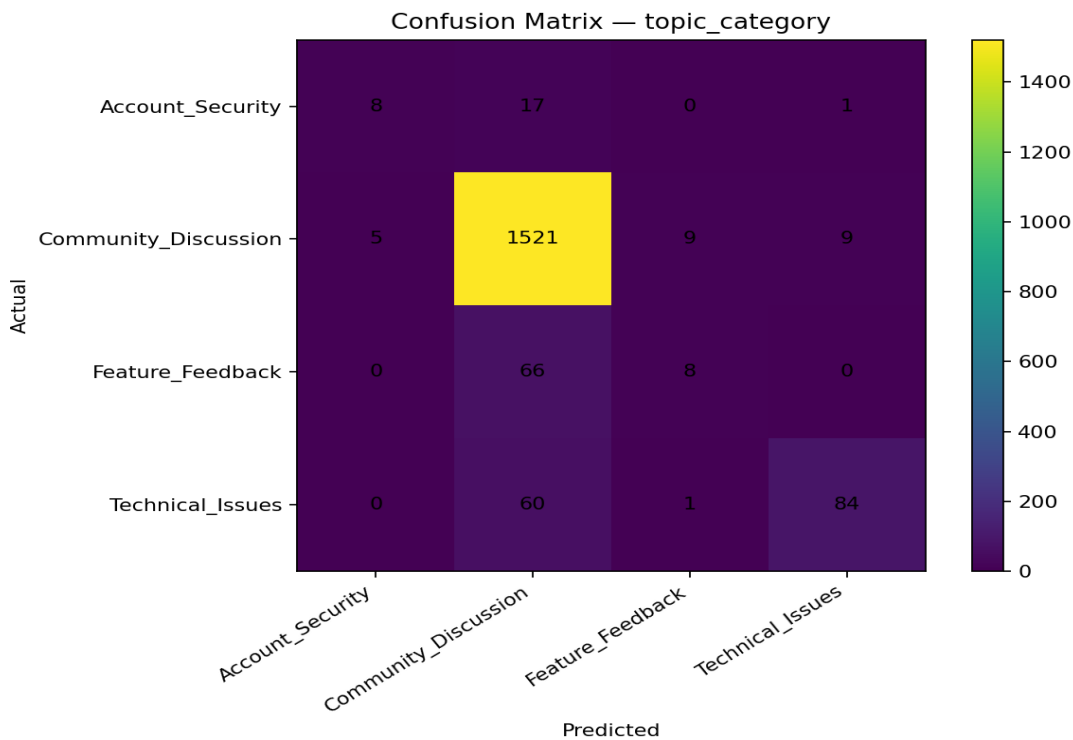


Figure 2. Topic confusion matrix (archived output).

5. Error Analysis & Interpretation

Sentiment: neutral sentiment remains the weakest class in the supplied improved report (F1 0.53), consistent with the difficulty of separating mixed, ambiguous, or context-dependent language. Inspect neutral-to-positive and neutral-to-negative confusions to validate this interpretation.

Topics: improved recall is 0.35 for Account_Security and 0.32 for Feature_Feedback, despite precision of 1.00 and 0.92. This indicates that the classifier identifies a relatively narrow subset of these classes and misses many true examples. Community_Discussion has F1 0.97, while Technical_Issues has F1 0.92. The archived training distribution was previously summarized as strongly imbalanced; verify the exact counts from the notebook/data before quoting them in a final submission.

6. Evaluation Notes & Limitations

- Baseline JSON metrics and improved reports are from different recorded outputs; confirm data split, model version, and preprocessing parity before claiming a strictly controlled comparison.
- Validation tuning results must not be presented as final test performance.
- No causal or real-world user-impact claim is inferred from classification metrics alone.
- Add [evaluation protocol / split details] if not documented in the notebook.

Appendix: archived files include sentiment/topic model pickle files, confusion-matrix images, topic error-analysis worksheet, and the Round 2 notebook.